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SCC WebApp Rebuild Plan

1. Project Objective

Main Goal

Rebuild the Ghana Supreme Court Cases WebApp into a modern, accessible, reliable and scalable platform for discovering Ghanaian Supreme Court cases.

The rebuilt platform should:

1. **Pull structured case metadata from Wikidata.**
2. **Retrieve associated images/files from Wikimedia Commons.**
3. **Connect cases to full judgment texts on Wikisource where available.**
4. **Provide powerful search and filtering.**
5. **Present case information in a clean, professional legal-research interface.**
6. **Clearly indicate when information is unavailable rather than displaying broken or misleading content.**
7. **Be responsive across mobile, tablet and desktop.**
8. **Remain open-source and aligned with the Wikimedia ecosystem.**

The current Wikidata records already contain useful fields such as case name, publication date, legal citation, court, judges and publication information.

2. Rebuild Philosophy

The rebuild should follow this principle:

Wikidata is the structured data layer. Wikimedia Commons is the media layer. Wikisource is the document/text layer. The SCC WebApp is the presentation and discovery layer.

This separation is important.

The application should **not become another independent database containing duplicated case information.**

Instead:

Wikidata → Case Metadata

↓

Wikimedia Commons → Images / Scanned Documents / Media

↓

Wikisource → Transcribed / Readable Judgment Text

↓

SCC WebApp → Search, Discovery, Presentation & User Experience

This keeps the project aligned with its original open-knowledge objective.

3. Phase 1 — Audit The Existing WebApp

Before writing new code, conduct a complete technical and functional audit.

A. Existing Interface Audit

Document every current page and feature:

- Homepage
- Search
- Case listing
- Case details
- Filters
- About page
- Navigation
- External links
- Language/translation functionality
- Mobile interface
- Error states
- Loading states
- Empty states

B. Backend Audit

Determine:

- Current programming language/framework
- Current API architecture

- SPARQL queries
- Wikidata properties being queried
- How results are cached
- How errors are handled
- How pagination works
- How filtering works
- How case URLs are generated
- How external Wikimedia links are generated

C. Wikimedia Integration Audit

Test every integration independently.

Wikidata

- Does the query return cases?
- Are all expected properties retrieved?
- Are dates correctly interpreted?
- Are judges correctly retrieved?
- Are citations correctly displayed?
- Are Wikidata QIDs retained?

Wikimedia Commons

- Does the app correctly identify associated files?
- Are thumbnails loading?
- Are file titles correct?
- Are attribution/licensing details available?
- What happens when no image exists?

Wikisource

- Does the app correctly identify available texts?
- Do links resolve?
- Can the application distinguish between a Wikisource page and a full document?
- What happens when a case has no Wikisource text?

4. Phase 2 — Define The Data Model

Before redesigning the frontend, establish exactly what constitutes a **Case**.

A case record should ideally contain:

Core Information

- Case name
- Wikidata Q-ID
- Publication date
- Decision date
- Legal citation
- Court
- Jurisdiction
- Case type
- Case number, where available

Parties

- Appellant
- Respondent
- Other parties
- Related persons/entities

Judicial Information

- Judges
- Presiding judge
- Majority opinion author, where available

Publications

- Ghana Law Reports citation
- Volume
- Page
- Publication information

Wikimedia Connections

- Wikidata URL
- Wikimedia Commons file/category
- Wikisource page
- Related Wikimedia resources

For example, current Wikidata entries can contain the case title, publication date, citation, court and judges.

5. Phase 3 — Rebuild The Data Pipeline

This should be one of the most important parts of the project.

Layer 1: Wikidata

Create a reliable query layer responsible for retrieving:

Case

- QID
- Title
- Date
- Citation
- Court
- Country
- Judges
- Parties
- Publication
- Related Wikimedia resources

The application should **never assume that every case contains every property**.

For example:

If a case has no judge → don't display "Unknown Judge" as if it were a confirmed fact.

Instead:

Judges: Not available

This distinction is important for legal information.

6. Wikimedia Commons Integration

The second layer should locate relevant Commons media.

Possible content:

- Supreme Court photographs
- Historical legal documents
- Scanned judgments

- Law reports
- Court photographs
- Other case-related media

The system should determine:

Does this case have associated Commons media?

If YES:

Display:

- Thumbnail
- File title
- Media type
- Source
- License information
- Link to Commons

If NO:

Use a clean placeholder.

Do **not** show broken image icons.

7. Wikisource Integration

This should be treated as a core feature rather than simply an external link.

For cases with a corresponding Wikisource text:

Case Page

Read Full Judgment

→ Opens the corresponding Wikisource text.

Where technically feasible, the rebuilt application could also provide a **reading experience inside the SCC interface**, while clearly identifying Wikisource as the source.

For example:

Full Judgment

Available on Wikisource

Then:

Read Judgment →

For cases without a Wikisource version:

Full Judgment

This case does not currently have a full text available on Wikisource.

This is much better than presenting an empty button.

8. New Homepage

The homepage should immediately explain what the platform does.

Hero Section

Ghana Supreme Court Cases

Explore structured information about Ghana's Supreme Court cases through an open, searchable platform powered by Wikimedia projects.

Then a prominent search bar:

Search cases, judges, citations or keywords

Example:

Search "Tetteh v Hayford"

Below it:

Browse By

- Year
- Judge
- Citation
- Case
- Legal topic
- Availability of full text

9. Search Experience

Search should become one of the strongest features.

Users should be able to search by:

Case Name

Tetteh v Hayford

Judge

Sophia Akuffo

Citation

[2012] GHASC 12

Year

2012

Keyword

constitutional

Wikidata ID

Q120197507

The search should provide suggestions while typing.

10. Advanced Filters

Add a dedicated filter panel.

Filters

Year

- 1960
- 1961
- 1962
- etc.

Judge

- Select judge

Court

- Supreme Court

Citation

- GLR
- SCGLR
- GHASC
- etc.

Full Text

- Available
- Not available

Media

- Commons available
- No media

Wikidata

- Complete
- Missing information

This could eventually become extremely useful for researchers.

11. Case Page Redesign

The case page should become the heart of the application.

Header

Tetteh and Another v. Hayford

[2012] GHASC 12

22 February 2012

Quick Information

Information	Details
Court	Supreme Court of Ghana
Date	22 February 2012
Citation	[2012] GHASC 12
Judges	Listed judges
Wikidata	Q-ID

Judges

Display judges as clickable entities where possible.

Documents

Full Judgment

Wikisource

Read the full judgment

Related Files

Wikimedia Commons

View available documents and media

Wikidata

A subtle “View on Wikidata” link should provide transparency about the source of the structured information.

12. Judge Pages

This could be a major improvement.

Instead of judges being plain text, create judge profiles.

For example:

Justice Sophia Akuffo

Cases in database: 124

Then:

- Cases
- Years served
- Decisions
- Related cases

This would allow researchers to explore the database through judges rather than only through case names.

13. Year-Based Exploration

Create a visual timeline.

Example:

1960 → 1970 → 1980 → 1990 → 2000 → 2010 → 2020 → Present

Selecting a year displays the available cases.

This could make the historical development of Ghana's Supreme Court much easier to explore.

14. Reliability & Error Handling

This needs to be built into the new application.

Instead of:

"Something went wrong."

Use specific messages.

Wikidata unavailable

We couldn't retrieve case information from Wikidata. Please try again.

Wikisource unavailable

The case information is available, but the full judgment could not be retrieved from Wikisource.

No Commons media

No related Wikimedia Commons media is currently available.

Missing Wikidata property

Citation not available in Wikidata.

This makes the application transparent.

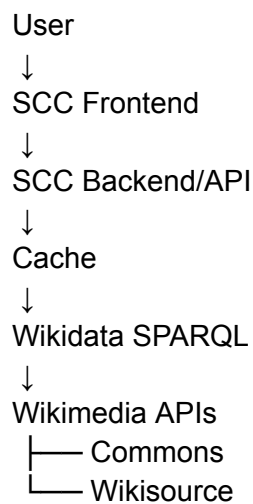
15. Caching & Performance

One major technical consideration is that the application should **not send unnecessary SPARQL queries every time a user loads a page**.

Implement:

- Query caching
- Pagination
- Lazy loading
- Request timeout handling
- Retry mechanisms
- API error handling

Potential architecture:



This will make the application faster and reduce unnecessary load on Wikimedia services.

16. Data Quality Dashboard

I strongly recommend adding an internal/admin-facing quality dashboard.

It could identify:

Missing Data

- Cases without citation
- Cases without dates
- Cases without judges
- Cases without Wikisource

- Cases without Commons media

Broken Connections

- Broken Wikisource links
- Missing Commons files
- Invalid Wikidata QIDs

Duplicate Cases

- Same case appearing multiple times
- Similar case names
- Conflicting dates

This is particularly important because the project is ultimately dependent on Wikidata quality.

17. Testing Strategy

Before launch, create a test dataset covering different periods and different levels of completeness.

For example:

Test Case A

Complete Wikidata + Commons + Wikisource.

Test Case B

Wikidata + Wikisource but no Commons.

Test Case C

Wikidata only.

Test Case D

Incomplete Wikidata.

Test Case E

Multiple judges.

Test Case F

Historical case.

Test Case G

Case with unusual citation formatting.

Every one should render correctly.

18. Automated Testing

The rebuilt application should include:

Unit Tests

Test:

- Data parsing
- Date conversion
- Citation formatting
- Judge extraction
- QID handling

Integration Tests

Test:

- Wikidata → application
- Commons → application
- Wikisource → application

End-to-End Tests

Test:

- Search
- Filters
- Case page
- External links

- Mobile navigation

Regression Tests

Every time the application is updated, automatically verify that the three Wikimedia integrations still work.

19. Accessibility

The new interface should follow modern accessibility principles.

Include:

- Keyboard navigation
- Proper heading structure
- Alt text
- Strong contrast
- Accessible forms
- Screen-reader-friendly navigation
- Visible focus states
- Responsive text
- Reduced-motion consideration

The application is intended for public use, so accessibility should be considered part of the core product rather than an optional enhancement.

20. Mobile-First Design

A significant portion of users may access the platform from phones.

Therefore:

Mobile

- Large search bar
- Simple navigation
- Collapsible filters
- Readable case information

- Easy “Read Judgment” button
- Fast loading

Desktop

- Sidebar filters
 - Larger result tables
 - More detailed case metadata
 - Multi-column layout
-

21. Proposed Navigation

A clean navigation could be:

Home

Cases

Judges

Browse

About

Contribute

With:

Search 

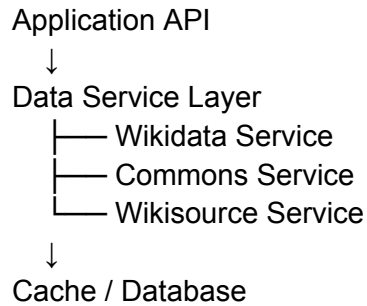
always accessible.

22. Suggested Technical Architecture

If the team is rebuilding rather than patching the existing application, I would recommend separating the application into:

Frontend





The important part is the **service layer**.

Instead of putting Wikimedia API calls directly throughout the frontend, create dedicated services:

WikidataService
CommonsService
WikisourceService

This makes the application much easier to maintain.

23. Project Phases

Phase 1 — Discovery & Audit

1 Week

- Audit current application
- Document existing functions
- Identify bugs
- Map Wikimedia integrations
- Define requirements
- Finalize new information architecture

Deliverable: Technical + UX Audit Report

Phase 2 — UI/UX Design

1–2 Weeks

- Wireframes
- Homepage
- Search
- Results
- Case page
- Judge page
- Mobile design
- Error states
- Loading states

Deliverable: Approved UI/UX prototype

Phase 3 — Data & API Architecture

- Wikidata queries
- Commons integration
- Wikisource integration
- Data normalization
- Caching
- Error handling

Deliverable: Working data layer

Phase 4 — Frontend Development

Build:

- Homepage
- Search
- Results
- Filters
- Case pages
- Judge pages
- Browse/timeline
- Responsive interface

Deliverable: Functional frontend

Phase 5 — Integration

Connect:

Frontend → **Backend** → **Wikidata** → **Commons** → **Wikisource**

Test real-world records.

Phase 6 — Quality Assurance

- Functional testing
 - API testing
 - Mobile testing
 - Accessibility testing
 - Performance testing
 - Broken-link testing
 - Data accuracy testing
-

Phase 7 — Deployment

- Production deployment
 - Monitoring
 - Error logging
 - Documentation
 - Developer handover
 - Contributor documentation
-

24. Definition Of Done

The SCC rebuild should **not** be considered complete simply because the new interface looks good.

The project is complete when:

Design

- ☐ Modern responsive interface
- ☐ Mobile optimized
- ☐ Desktop optimized
- ☐ Accessible

Wikidata

- ☐ Case data loads correctly
- ☐ Dates display correctly
- ☐ Judges display correctly
- ☐ Citations display correctly
- ☐ Missing fields handled gracefully
- ☐ Search works
- ☐ Filters work

Wikimedia Commons

- ☐ Files are correctly identified
- ☐ Images load
- ☐ Thumbnails work
- ☐ Links work
- ☐ Missing media handled correctly

Wikisource

- ☐ Available judgments are detected
- ☐ Links work
- ☐ Full-text availability is clearly indicated
- ☐ Missing texts handled gracefully

Performance

- ☐ Pages load quickly
- ☐ Queries are cached
- ☐ Pagination works
- ☐ API failures don't crash the application

Quality

- ☐ Automated tests
- ☐ Integration tests
- ☐ Mobile testing
- ☐ Accessibility testing
- ☐ Security review

- ☐ Documentation

Harriet

SCC WebApp Rebuild Plan

The proposed project is to rebuild and improve the existing Ghana Supreme Court Cases (SCC) WebApp to make it more modern, user-friendly, reliable and useful for researchers, law students, legal professionals and the general public.

The rebuild will focus on:

1. **Redesign & User Experience**

Improve the overall look and feel of the platform, making it more professional, appealing and easy to navigate on both mobile and desktop.

2. **Search & Discovery**

Improve the search functionality and introduce better filters for finding cases by year, judge, citation, case name and other relevant information.

3. **Wikimedia Integration**

Ensure that the platform correctly retrieves structured case information from **Wikidata**, related files and images from **Wikimedia Commons**, and available full judgment texts from **Wikisource**.

4. **Data Reliability**

Fix existing data and integration issues, improve how missing information is handled, and ensure that users are not presented with broken links or incomplete features.

5. **Testing & Deployment**

Thoroughly test the platform, including its search, data retrieval, Wikimedia integrations, mobile responsiveness and overall performance before deploying the improved version.

Expected Outcome

At the end of the project, we will have a modern and reliable SCC WebApp that provides a better way for the public and researchers to discover and access Ghana's Supreme Court cases through the Wikimedia ecosystem.

Proposed Timeline: 4 days

Key Deliverables:

- Redesigned SCC WebApp
- Improved search and filtering
- Functional Wikidata integration
- Functional Wikimedia Commons integration
- Functional Wikisource integration
- Mobile-responsive interface
- Improved performance and reliability
- Testing and final deployment